# LOW LATENCY AND EFFICIENT VOICE ACTIVATOR FOR EDGE DEVICES

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# MALLA REDDY UNIVERSITY

(Telangana State Private Universities Act No.13 of 2020 and G.O.Ms.No.14, Higher Education (UE) Department)

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## CERTIFICATE

This is to certify that this is the project report entitled **LOW LATENCY AND EFFICIENT VOICE ACTIVATOR FOR EDGE DEVICE" submitted by P SAI VARDHAN (2411EC010040) SYED AKHIF (2411EC010056) K SNEHAL (2411EC010028) A.AKHILA (2411EC010001) D.DEVI SRI GANGA (2411EC010013** B.TECH III year I semester, Department of Electronics and Communication Engineering(ECE),during the year 2026-27.the result embodied in this report have not been submitted to other university or institute for the award of any degree or diploma.

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# DECLARATION

We hereby declare that the project report “LOW LATENCY AND EFFICIENT VOICE ACTIVATOR FOR EDGE DEVICE” has been carried out by us and this work has been submitted to the Department of Electronics and Communication Engineering (ECE),Malla Reddy University,Hyderabad.We further declare that this project work has not been submittrd in full or part for the award of any other degree in any other educational institutions.

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# ABSTRACT

The Low-Latency and Efficient Voice Activator for Edge Devices is a voice-based activation system designed to detect a predefined wake word or voice command locally on an edge device. Unlike conventional voice assistants that depend heavily on cloud servers, the proposed system performs the initial voice detection and processing directly on the device.

This reduces response latency, minimizes network dependency, and improves privacy. The system continuously monitors audio input through a microphone using a lightweight voice-processing algorithm. When the specified activation word is detected, the system triggers the required action or passes the command to the next processing stage. The system

is designed to consume low computational power and memory, making it suitable for resource-constrained devices such as microcontrollers, embedded systems, smart appliances,

and IoT devices.

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# CHAPTER - 1 INTRODUCTION

The development of voice-controlled technology has become an important part of modern electronics and embedded systems. Voice activation allows users to interact with electronic devices through spoken commands without requiring physical contact. Traditional voice recognition systems often depend on cloud servers to process audio data, which can introduce communication delays, require continuous internet connectivity, and raise privacy concerns. To overcome these limitations, a Low Latency and Efficient Voice Activator for Edge Devices provides a solution by processing voice commands directly on local hardware.

Edge computing is a technology that performs data processing near the source of data instead of sending it to a remote server. In this project, an edge device, such as an ESP32 or ESP32-S3 microcontroller, is used to capture audio signals, detect voice activation commands, and trigger specific actions. The system can be designed using a microphone, a microcontroller, and an output device such as an LED or OLED display to indicate successful voice detection.

The primary objective of this project is to reduce response time while maintaining efficient utilization of memory, processing power, and energy. By performing voice detection locally, the system minimizes network dependency and improves reliability. Lightweight algorithms and optimized signal-processing techniques can help the device recognize predefined keywords with limited computational resources. This approach is particularly useful for embedded applications where low power consumption and fast responses are essential.

The proposed system has potential applications in smart homes, industrial automation, assistive technologies, healthcare devices, and intelligent robotics. It can enable hands-free operation, improve accessibility, and support real-time control in environments where internet connectivity is limited or unavailable.

In conclusion, a low-latency and efficient voice activator for edge devices demonstrates how embedded systems and voice-processing technologies can be combined to develop responsive, reliable, and cost-effective solutions. The project focuses on achieving faster voice activation, reduced computational overhead, improved privacy, and efficient local processing, making it a promising approach for next-generation intelligent edge applications.

## 1.1 Background of the Problem

The rapid development of embedded systems, artificial intelligence, and edge computing has increased the demand for intelligent voice-controlled devices. Voice activation technology enables users to operate electronic systems through spoken commands, making human-machine interaction easier and more convenient. Traditional voice recognition systems often rely on cloud-based processing, which requires internet connectivity and may introduce delays due to data transmission. To overcome these challenges, edge computing enables voice processing directly on local hardware, such as the ESP32 or ESP32-S3 microcontroller. These devices can perform basic audio processing and keyword detection with low power consumption and limited computational resources. By processing voice commands locally, the system can reduce response time, improve privacy, and operate with less dependence on external servers.

The Low Latency and Efficient Voice Activator for Edge Devices project focuses on developing a compact, responsive, and energy-efficient voice activation system. It aims to demonstrate how lightweight voice detection techniques can be implemented in embedded devices for applications such as smart homes, robotics, industrial automation, and assistive technologies.

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## 1.2 Motivation and Need for the Project

The increasing use of smart devices, automation systems, and artificial intelligence has created a growing need for fast, reliable, and efficient voice-controlled technologies. Voice activation allows users to control electronic devices through spoken commands, reducing the need for physical switches and manual operation. However, many traditional voice recognition systems depend on cloud servers, which require internet connectivity and may introduce delays during communication and processing.

The main motivation behind the Low Latency and Efficient Voice Activator for Edge Devices project is to overcome these limitations by performing voice detection directly on edge hardware. By using a microcontroller such as the ESP32 or ESP32-S3, the system can process predefined voice commands locally, reduce response time, and minimize dependence on external servers. Local processing also improves privacy by reducing the need to transmit audio data over the internet.

This project is needed to develop an affordable, compact, and energy-efficient voice activation system suitable for embedded applications. It aims to optimize processing time and resource utilization while maintaining reliable voice detection. The system can be applied in smart home automation, robotics, industrial control, and assistive devices for people with limited mobility. Overall, the project promotes faster human-machine interaction and demonstrates the practical benefits of combining embedded systems, voice processing, and edge computing to develop intelligent and responsive electronic devices.

It helps students and developers learn how hardware and software work together to create intelligent behaviors in machines. Additionally, the project focuses on using affordable components like Arduino, Bluetooth modules, and open-source libraries, making it budget-friendly and easy to replicate for educational or practical purposes. The demand for a voice-controlled AI robot stems from the need for more natural, efficient, and intelligent ways for humans to interact with machines. It fills the gap between virtual assistants that can only speak and physical robots that can act but lack intelligence. By combining these functions, the system offers a more complete form of assistance—one that can listen, understand, and perform tasks. In conclusion, the project aims to create a personal assistant that improves human comfort, supports accessibility, and showcases the practical power of AI and robotics. It represents a move toward a future where technology not only follows commands but also interacts intelligently and empathetically with humans.

### 1.3 Problem Statement

Traditional voice activation systems often depend on cloud-based servers for voice processing and command recognition, resulting in increased response time, internet dependency, higher communication overhead, and potential privacy concerns. These limitations make such systems less suitable for real-time applications and resource- constrained embedded devices.

The Low Latency and Efficient Voice Activator for Edge Devices project aims to develop a voice activation system that detects predefined voice commands locally using an edge device, such as the ESP32 or ESP32-S3. The system focuses on minimizing response latency, reducing computational and energy requirements, and improving reliability without depending on continuous internet connectivity. The proposed solution seeks to provide fast, cost-effective, and efficient voice control for applications such as robotics, smart home automation, industrial control, and assistive technologies.

## 1.4 Objectives of the Project

- 
- The main objective of the Low Latency and Efficient Voice Activator for Edge Devices
- project is to develop a fast, reliable, and energy-efficient voice activation system that
- processes predefined voice commands directly on edge hardware.
- The specific objectives of the project are:

  1. Reduce Latency: Minimize the time between detecting a voice command and generating the corresponding output.

  2. Local Voice Processing: Implement voice activation on an edge device, such as the ESP32-S3, to reduce dependence on cloud-based servers.

  3. Improve Efficiency: Optimize memory usage, processing power, and energy consumption for resource-constrained embedded systems.

  4. Enable Real-Time Response: Detect predefined voice commands and trigger appropriate actions with minimal delay.

  5. Enhance Privacy: Reduce the need to transmit audio data to external servers by performing voice detection locally.

  6. Develop a Low-Cost System: Use affordable hardware components, such as a microcontroller, microphone, and OLED display, to build a compact prototype.

  7. Demonstrate Practical Applications: Evaluate the system's suitability for robotics, smart home automation, industrial control, and assistive technologies.

  8. Evaluate Performance: Assess voice detection accuracy, response time, reliability, and resource utilization under different operating conditions.

  These objectives aim to demonstrate the feasibility of implementing an efficient, low-latency voice activation system for modern edge computing applications.

## 1.5 Scope of the Project

The scope of the Low Latency and Efficient Voice Activator for Edge Devices project is to design and develop a compact, cost-effective, and energy-efficient voice activation system using edge computing technology. The system uses a microcontroller such as the ESP32-S3 to process voice inputs and detect predefined commands locally, reducing dependence on cloud- based processing and continuous internet connectivity.

The project includes hardware integration, voice input acquisition, command detection, response generation, and output indication using components such as a microphone, microcontroller, LED, or OLED display. It focuses on reducing response latency, optimizing computational resources, and improving system reliability.

The proposed system can be extended to applications such as smart home automation, voice-controlled robotics, industrial equipment monitoring, and assistive devices. It also provides opportunities to integrate lightweight machine learning models for improved keyword detection and recognition accuracy.

The performance of the system can be evaluated based on response time, detection accuracy, power consumption, and memory utilization. However, the recognition capability depends on the hardware, microphone quality, background noise, and voice detection algorithm used. Overall, this project provides a foundation for developing intelligent, responsive, and privacy-conscious voice-controlled edge devices.

## 1.6 Organization of the Report

- This project report is organized into different chapters to
- explain the design, development, implementation, and
- evaluation of the Low Latency and Efficient Voice
- Activator for Edge Devices.
- Chapter 1: Introduction introduces the project, including
- the background, motivation and need, problem statement, objectives, scope, and organization of the report.

  Chapter 2: Literature Review discusses existing voice activation systems, edge computing technologies, voice detection techniques, and related research. It also identifies the limitations of existing approaches.

  Chapter 3: System Requirements and Architecture describes the hardware and software requirements, including the ESP32-S3 microcontroller, microphone, OLED display, and Arduino IDE. It also presents the proposed system architecture and block diagram.

  Chapter 4: System Design and Methodology explains the hardware connections, software design, voice processing approach, command detection, and overall working procedure.

  Chapter 5: Implementation and Testing presents the implementation process, program development, hardware integration, and testing procedures used to verify the system's functionality.

  Chapter 6: Results and Performance Analysis evaluates the system based on response latency, voice detection accuracy, power consumption, and resource utilization.

  Chapter 7: Conclusion and Future Scope summarizes the project outcomes, discusses its limitations, and suggests future improvements, such as advanced keyword recognition and integration with additional automation systems.

  References: This section lists the research papers, technical documentation, books, and other resources used during the project.

# CHAPTER - 2

## Literature Review and Existing System

### 2.1 Overview of Previous Works

Voice activation technology has developed significantly with advances in speech recognition, embedded systems, artificial intelligence, and edge computing. Early voice recognition systems relied on computers with high processing capabilities to identify spoken commands. These systems often required considerable computational resources and were not suitable for small, low-power embedded devices.

Later, cloud-based voice assistants introduced more advanced speech recognition capabilities. These systems process audio data on remote servers and provide responses to user commands. Although they offer improved recognition performance, they depend on internet connectivity and may experience delays due to network communication. They can also raise privacy concerns because voice data may be transmitted to external servers.

Recent research has focused on edge-based voice processing using microcontrollers and lightweight machine learning algorithms. Platforms such as ESP32 and ESP32-S3 support embedded applications that can perform predefined keyword detection locally. Techniques such as digital signal processing, feature extraction, and compact neural networks help reduce computational complexity and response time.

Existing studies demonstrate that local voice processing can improve responsiveness, reduce network dependency, and support privacy-conscious applications. However, challengesremain in achieving high detection accuracy under noisy conditions while maintaining low power consumption and limited memory usage.

Therefore, the proposed Low Latency and Efficient Voice Activator for Edge Devices project focuses on developing an affordable embedded system that detects predefined voice commands locally and provides fast responses with efficient resource utilization.

## 2.2 Limitations in Existing Systems

Existing voice activation systems provide convenient hands-free control, but they have several limitations related to response time, power consumption, privacy, and computational requirements. These limitations create challenges when implementing voice activation on resource-constrained edge devices.

1. High Response Latency: Cloud-based voice recognition systems require audio data to be transmitted to remote servers, processed, and returned to the device. Network delays can increase response time.
2. Internet Dependency: Many traditional voice assistants require a stable internet connection to recognize commands and perform operations, limiting their functionality in areas with poor connectivity.
3. Privacy Concerns: Transmitting voice recordings to external servers may create data privacy and security risks.
4. High Computational Requirements: Advanced speech recognition models often require significant processing power, memory, and storage, making them difficult to implement on low-cost microcontrollers.
5. Power Consumption: Continuous audio processing and communication with remote servers can increase energy consumption, particularly in battery-powered devices.
6. Background Noise: Environmental sounds, varying accents, and differences in speaking volume can reduce voice detection accuracy.
7. Limited Hardware Resources: Small embedded devices have restricted RAM, flash storage, and processing capabilities, which limit the complexity of voice recognition algorithms.
8. System Reliability: Network failures, microphone quality, and software limitations can affect consistent operation.

These limitations highlight the need for a low-latency, energy-efficient, and locally operating voice activation system that can deliver reliable performance with reduced internet dependency and computational overhead.

## 2.3 Research Gap Identified

Existing voice activation systems have achieved significant progress in speech recognition and human-machine interaction. However, several challenges remain in developing efficient voice activation systems for resource-constrained edge devices. Cloud-based systems often depend on stable internet connectivity and may introduce additional latency due to remote processing. Although edge-based systems reduce network dependency, they must operate within limited memory, processing power, and energy budgets.

A major research gap is the need to balance low response latency, voice detection accuracy, and computational efficiency within a single embedded system. Some existing solutions focus primarily on recognition accuracy, while others prioritize reduced power consumption or faster processing. Achieving all these requirements simultaneously remains challenging, particularly in noisy environments and on low-cost microcontrollers.

Furthermore, voice detection performance can vary depending on microphone quality, background noise, speaking patterns, and the algorithm used. There is a need to evaluate system performance using measurable parameters such as response time, detection accuracy, memory utilization, and power consumption.

The proposed Low Latency and Efficient Voice Activator for Edge Devices project addresses these challenges by implementing local voice command detection using an edge device such as the ESP32-S3. It aims to minimize processing delays, reduce dependence on cloud services, and optimize hardware resource utilization. The project provides a foundation for developing affordable and responsive voice-controlled applications in robotics, smart home automation, and other embedded systems.

## 2.4 Summary

This chapter presented an overview of existing voice activation systems, their development, limitations, and the research gap identified in current approaches. Voice activation technology has become an important part of modern electronics, embedded systems, and human-machine interaction. Traditional voice recognition systems generally depend on computers or cloud-based servers to process audio data and identify spoken commands. Although these systems provide advanced recognition capabilities, they may experience delays due to network communication and require continuous internet connectivity.

The literature review highlighted the growing importance of edge computing for developing faster and more efficient voice-controlled applications. Edge devices process data locally, reducing the need to transmit audio information to external servers. Microcontrollers such as the ESP32 and ESP32-S3 provide suitable platforms for implementing lightweight voice activation systems because of their processing capabilities, compact size, and support for embedded applications. Several limitations were identified in existing systems, including high response latency, power consumption, privacy concerns, limited memory, computational complexity, and reduced recognition accuracy in noisy environments. These challenges indicate the need for a system that balances processing speed, detection accuracy, energy efficiency, and hardware resource utilization. The research gap identified in this chapter focuses on developing a cost-effective voice activation solution capable of operating efficiently on resource-constrained edge devices.

The proposed Low Latency and Efficient Voice Activator for Edge Devices project aims to address these challenges by implementing local voice command detection and generating appropriate outputs with minimal delay. The system uses an edge computing platform, a microphone, and output indicators such as an LED or OLED display to demonstrate voice activation. Its performance can be evaluated using parameters such as response time, detection accuracy, memory utilization, and power consumption.

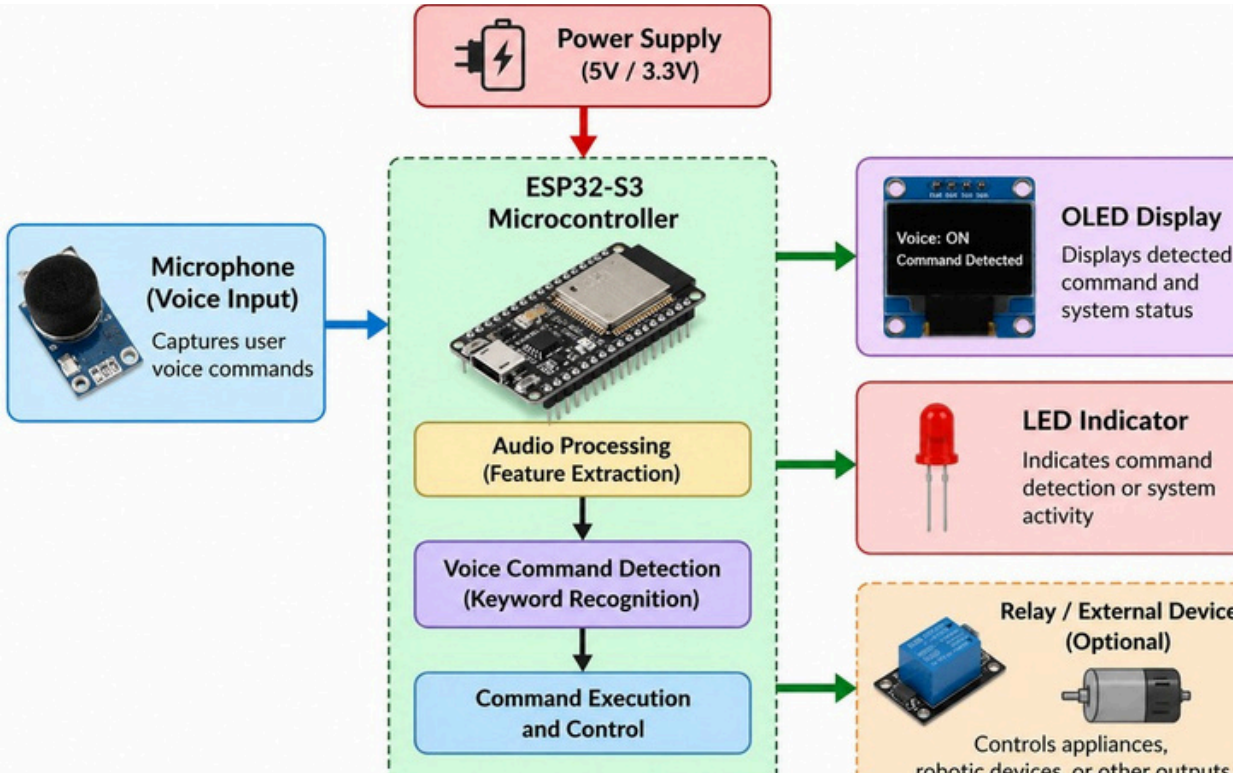
In conclusion, this chapter establishes the foundation for the proposed system by identifying the limitations of existing approaches and explaining the need for efficient local voice processing. The next chapter discusses the system requirements, hardware components, software tools, and proposed architecture required for implementation.

CHAPTER - 3

Proposed System / Methodology

## 3.1 System Architecture / Block Diagram

The system architecture of the Low Latency and Efficient Voice Activator for Edge Devices illustrates the interaction between the hardware and software components used to detect voice commands and generate appropriate outputs. The proposed system uses an ESP32-S3 microcontroller to process voice input locally, reducing dependence on cloud servers and minimizing communication delays.



### 3.2 Working Principle

1. Voice Input: The user speaks a predefined keyword or command.
2. Audio Acquisition: The microphone captures the voice signal and provides it to the processing system.
3. Local Processing: The ESP32-S3 processes the audio using the selected voice detection algorithm or lightweight model.
4. Command Detection: The system determines whether the input matches a supported command.
5. Output Generation: When a command is detected, the OLED displays the result and an LED or another output device indicates activation.

Key Design Features

- Low latency: Local processing reduces network-related delays.
- Energy efficiency: Lightweight processing helps limit resource consumption.
- Privacy: Voice data can be processed locally without sending it to a cloud server.
- Compact design: The system can be implemented using a microcontroller and a small number of components.

Note: The actual performance depends on the microphone interface, voice detection algorithm, and hardware configuration. The ESP32-S3 does not automatically recognize arbitrary speech; a compatible keyword detection model or voice recognition implementation is required.

### 3.3 Detailed Explanation of Each Block

- The system consists of the following blocks:
- 1.Power Supply: Provides the required regulated voltage to the
- microcontroller and connected components.
- 2.Microphone: Captures the user's voice commands and converts sound into
- electrical or digital signals.

3.ESP32-S3 Microcontroller: Processes audio data and controls the overall system.

4.Audio Processing: Filters noise and extracts useful voice features for detection.

5.Voice Command Detection: Identifies predefined keywords using a compatible voice detection algorithm.

6.Command Execution: Performs the action associated with the recognized command.

7.OLED Display: Displays system status and detected commands.

8.LED Indicator: Provides visual feedback when a command is detected.

9.Optional Output: Controls external devices through a suitable driver or relay module.

### 3.4 Flowchart / Algorithm Steps

1.Start the system.
2.Initialize all modules (Bluetooth, sensors, motors).
3.Wait for voice input.
4.Capture and process the command.
5.Match the command with predefined instructions.
6.Execute corresponding action (move, stop, speak, etc.). 7.Provide feedback to the user.
8.Repeat until system shutdown.

### 3.5 Hardware and SoftwareRequirements

Hardware:

- ESP32-S3 Development Board
- I²S Microphone Module (INMP441)
- 0.96-inch OLED Display (SSD1306)
- LED
- 220–330 Ω Resistor
- Breadboard
- Jumper Wires
- USB Data Cable
- USB Power Supply

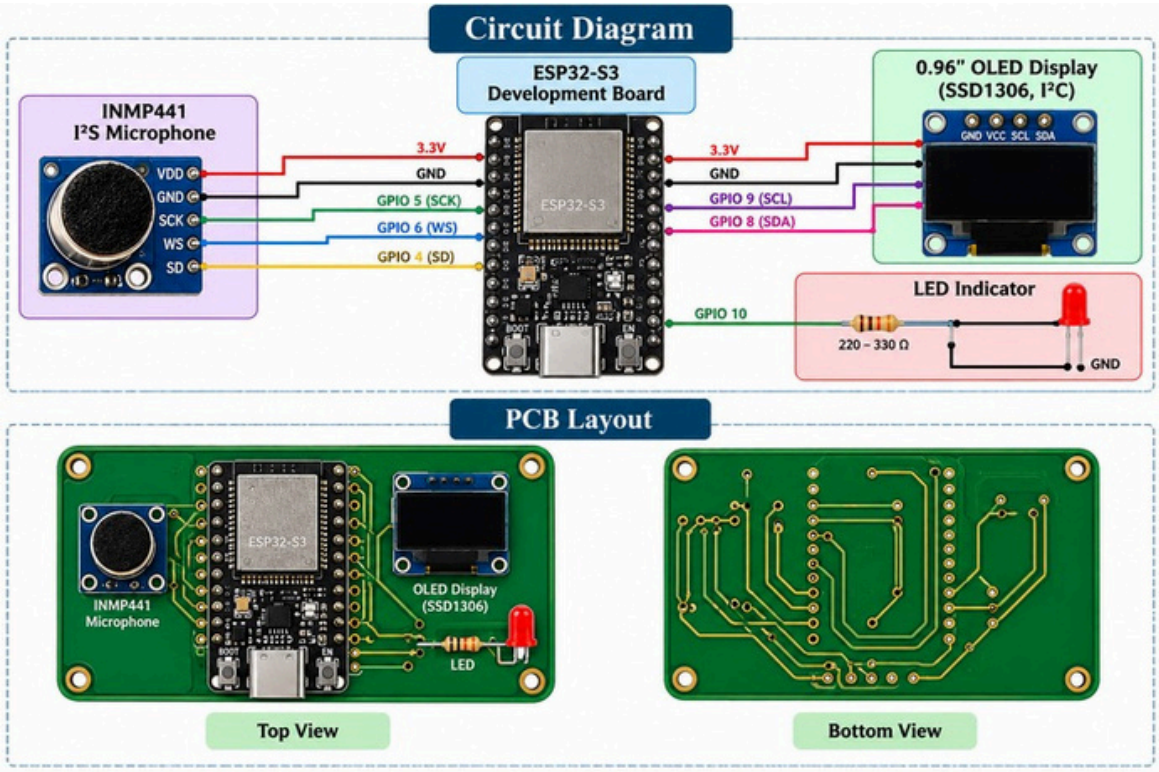
Software:

1. Arduino IDE
2. ESP32 Board Package
3. C/C++ Programming Language
4. Adafruit SSD1306 Library
5. Adafruit GFX Library
6. ESP32 I²S Audio Driver
7. TensorFlow Lite for Microcontrollers (if using a compatible machine-learning model)

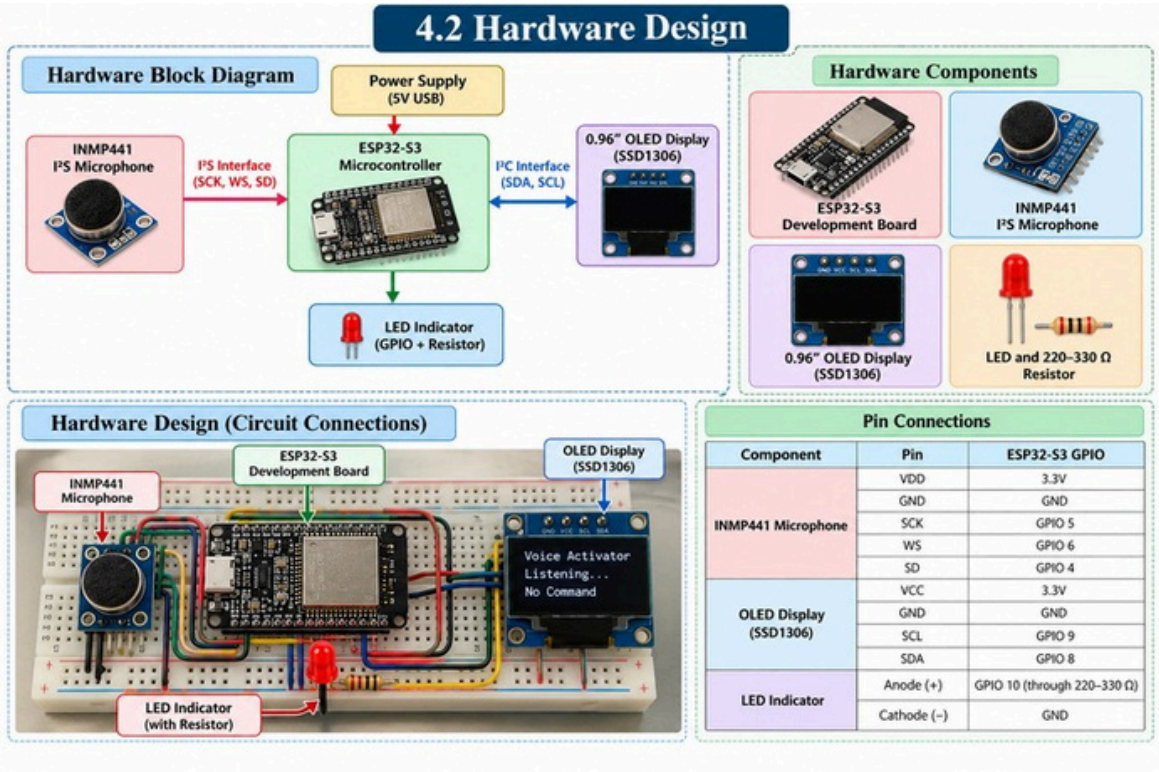
# CHAPTER - 4

# Design and Implementation

## 4.1 Circuit Diagrams / PCB Layouts



## 4.2 Hardware Design




| Component | Pin | ESP32-S3 GPIO |
|---|---|---|
| INMP441 Microphone | VDD | 3.3V |
| | GND | GND |
| | SCK | GPIO 5 |
| | WS | GPIO 6 |
| | SD | GPIO 4 |
| OLED Display (SSD1306) | VCC | 3.3V |
| | GND | GND |
| | SCL | GPIO 9 |
| | SDA | GPIO 8 |
| LED Indicator | Anode (+) | GPIO 10 (through 220–330 Ω) |
| | Cathode (–) | GND |

**4.3 Software Design**

The software design of the Low Latency and Efficient Voice Activator for Edge Devices focuses on developing a program that captures voice input, detects predefined keywords, and generates the appropriate output using the ESP32-S3 microcontroller.

The software is developed using the Arduino IDE and C/C++ programming language. It integrates the microphone, OLED display, and LED indicator through suitable libraries and communication interfaces.

The main software modules are:

1. Initialization Module: Configures the ESP32-S3, microphone, OLED display, and GPIO pins.
2. Audio Acquisition Module: Receives digital audio samples from the INMP441 microphone using the $I^2S$ interface.
3. Audio Processing Module: Prepares audio samples for voice detection using suitable signal-processing techniques.
4. Voice Detection Module: Identifies predefined keywords using a compatible keyword spotting algorithm or machine-learning model.
5. Output Control Module: Activates the LED and updates the OLED display when a valid command is detected.
6. Monitoring Module: Continuously monitors audio input and repeats the detection process.

The software follows a continuous execution loop in which audio is captured, processed, analyzed, and checked against predefined commands. Local processing reduces network dependency and can minimize response latency.

The overall software design aims to provide reliable voice activation, efficient resource utilization, and fast response on an embedded edge device. Actual detection performance depends on the selected voice recognition algorithm and its compatibility with the ESP32-S3.



## CODE AND IMPLEMENTATION

```
#include <Arduino.h>
#include <Wire.h>
#include <math.h>
#include <driver/i2s.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>

// OLED pins
#define OLED_SDA 8
#define OLED_SCL 9

#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
#define OLED_ADDRESS 0x3C

Adafruit_SSD1306 display(
 SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1
);

// INMP441 microphone pins
#define I2S_PORT I2S_NUM_0
#define I2S_BCLK 4
#define I2S_WS 5
#define I2S_DIN 6

#define SAMPLE_RATE 16000
#define BUFFER_SIZE 512

// Adjust after checking Serial Monitor readings
#define SOUND_THRESHOLD 500

int32_t audioBuffer[BUFFER_SIZE];

uint32_t audioLevel = 0;
bool soundDetected = false;

unsigned long soundUntil = 0;
unsigned long lastDisplayUpdate = 0;

void showStatus(const char *status) {
  display.clearDisplay();
  display.setTextColor(SSD1306_WHITE);
  display.setTextSize(1);
```

```
display.setCursor(0, 0);
 display.println("VOICE ACTIVATOR");
display.println("----------------");
 display.setCursor(0, 22);
 display.print("Status: ");
 display.println(status);
 display.setCursor(0, 42);
 display.print("Audio level: ");
 display.println(audioLevel);
 display.display();
}
bool startMicrophone() {
 i2s_config_t config = {
 .mode = (i2s_mode_t)(
 I2S_MODE_MASTER | I2S_MODE_RX
 ),
 .sample_rate = SAMPLE_RATE,
 .bits_per_sample = I2S_BITS_PER_SAMPLE_32BIT,
.channel_format = I2S_CHANNEL_FMT_ONLY_LEFT,
.communication_format = I2S_COMM_FORMAT_STAND_I2S,
.intr_alloc_flags = ESP_INTR_FLAG_LEVEL1,
.dma_buf_count = 4,
 .dma_buf_len = BUFFER_SIZE,
 .use_apll = false,
 .tx_desc_auto_clear = false,
 .fixed_mclk = 0
 };
 i2s_pin_config_t pins = {
 .bck_io_num = I2S_BCLK,
 .ws_io_num = I2S_WS,
 .data_out_num = I2S_PIN_NO_CHANGE,
 .data_in_num = I2S_DIN
 };
 if (i2s_driver_install(
 I2S_PORT, &config, 0, nullptr) != ESP_OK) {
 return false;
 }
 if (i2s_set_pin(I2S_PORT, &pins) != ESP_OK) {
i2s_driver_uninstall(I2S_PORT);
 return false;
 }
 i2s_zero_dma_buffer(I2S_PORT);
 return true;
}
```

```
void setup() {
 Serial.begin(115200);
 delay(500);
 // Start OLED
 Wire.begin(OLED_SDA, OLED_SCL);
 if (!display.begin(
 SSD1306_SWITCHCAPVCC, OLED_ADDRESS)) {
Serial.println("OLED not found!");
 while (true) {
 delay(1000);
 }
 }
 showStatus("STARTING");
 // Start microphone
 if (!startMicrophone()) {
 showStatus("MIC ERROR");
 Serial.println("I2S setup failed!");
 while (true) {
 delay(1000);
 }
 }
                              showStatus("LISTENING");
Serial.println("Voice activator ready.");
Serial.println("Speak near the microphone.");
}
void loop() {
 size_t bytesRead = 0;
 esp_err_t result = i2s_read(
 I2S_PORT,
 audioBuffer,
 sizeof(audioBuffer),
 &bytesRead,
 portMAX_DELAY
 );
 if (result != ESP_OK || bytesRead == 0) {
 return;
 }
  int samples = bytesRead / sizeof(int32_t);
uint64_t energy = 0;
 for (int i = 0; i < samples; i++) {
 // Convert the microphone's 24-bit
 // signal in a 32-bit I2S slot.
 int32_t sample = audioBuffer[i] >> 14;
 energy += (uint64_t)(
 (int64_t)sample * sample
 )
```

```
 }
 if (samples > 0) {
 audioLevel = (uint32_t)sqrt(
(double)energy / samples
 );
 }
 // Simple sound-activity detection
 if (audioLevel > SOUND_THRESHOLD) {
soundDetected = true;
 soundUntil = millis() + 1000;
Serial.print("Sound detected: ");
Serial.println(audioLevel);
 }
 if (millis() > soundUntil) {
soundDetected = false;
 }
 // Refresh OLED every 200 ms
 if (millis() - lastDisplayUpdate >= 200) {
lastDisplayUpdate = millis();
 if (soundDetected) {
showStatus("SOUND DETECTED");
 } else {
 showStatus("LISTENING...");
 }
 Serial.print("Audio level: ");
Serial.println(audioLevel)
```

# CODE FLOW

**START**
**Initialize ESP32-S3, I²S microphone, OLED and LED**
**Configure audio input and voice detection model**
**Capture audio samples from microphone**
**Process audio and extract features**
**Voice command detected?**
**YES**
**Activate LED / Execute action**
**Display command on OLED**
**NO**
**Continue listening**
**Capture next audio samples**
**Repeat loop until stopped**

**Code Logic Explanation**

The code logic of the Low Latency and Efficient Voice Activator for Edge Devices describes how the ESP32-S3 processes voice input and controls the output devices.

1. Initialization: The program initializes the ESP32-S3, I²S microphone, OLED display, and LED indicator.
2. Audio Acquisition: The microphone captures the user's voice and sends digital audio samples to the microcontroller through the I²S interface.
3. Audio Processing: The captured audio samples are processed and converted into suitable input features for the voice detection algorithm.
4. Voice Detection: A compatible keyword spotting model analyzes the audio and identifies predefined voice commands.
5. Decision Making: The program checks whether the detected keyword matches a supported command.
6. Output Control: If a valid command is detected, the LED is activated and the OLED displays the corresponding message. Otherwise, the system continues listening.
7. Continuous Operation: The program repeats the audio capture and detection process until the device is switched off.

This logic enables local voice processing, reduces dependence on internet connectivity, and supports fast responses. Actual performance depends on the voice detection model, microphone configuration, and available processing resources.

## 4.4 Communication Protocols Used

- TheLow Latency and Efficient VoiceActivator for Edge Devices project uses communication protocols to transfer audio data and display system information between the ESP32-S3 microcontroller and
- peripheral components.
- 1. I²S (Inter-IC Sound)
  I²S is used to transfer digital audio data from the INMP441 microphone to the ESP32-S3. It uses signals such as Serial Clock (SCK), Word Select (WS), and Serial Data (SD). This interface enables continuous
- audio sampling for voice command detection.
- 2. I²C (Inter-Integrated Circuit)
- I²C is used for communication between the ESP32-S3 and the SSD1306 OLED display. It uses two signal lines: Serial Data (SDA) and Serial Clock (SCL). The display shows messages such as system status and detected voice commands.
- 3. GPIO (General-Purpose Input/Output)
- GPIO pins are used to control the LED indicator and other output devices. When a valid voice command is detected, the ESP32-S3 changes the corresponding output pin state to perform the required action. 4. USB Serial Communication
- USB serial communi cation is used to upload the program from the Arduino IDE and monitor debugging messages through the Serial Monitor. It helps developers test and troubleshoot the system.
- .

**44.5 System Integration Details**

System integration involves connecting the hardware and software components of the Low Latency and Efficient Voice Activator for Edge Devices to operate as a unified system. The ESP32-S3 acts as the central controller, integrating the microphone, OLED display, and LED indicator.

1. Hardware Integration: The INMP441 microphone is connected to the ESP32-S3 through the $I^2S$ interface to capture voice input. The SSD1306 OLED display is connected through the $I^2C$ interface to display system status. The LED is connected to a GPIO pin through a current-limiting resistor.
2. Software Integration: The program is developed using the Arduino IDE. It initializes the connected components, acquires audio samples, processes voice input, and detects predefined commands using a compatible voice detection algorithm.
3. Functional Integration: When a valid command is detected, the ESP32-S3 performs the corresponding action and updates the OLED display or LED indicator. Otherwise, the system continues listening for new commands.
4. Testing and Verification: Each component is tested individually before integrating the complete system. The integrated prototype is then evaluated for voice detection accuracy, response latency, and reliable operation.

Successful integration enables the system to perform local voice detection with reduced dependence on internet connectivity and efficient use of embedded resources.

**Communication and Synchronization**

Communicationand synchronization are important for ensuring the proper coordination of hardware andsoftware components in the Low Latency and Efficient Voice Activator for Edge Devices project.

The ESP32-S3communicates with the INMP441 microphone through the $I^2S$ protocol to receive digital audio samples. The $I^2S$ clock signals synchronize audio data transmission and help maintain a consistent sampling rate. The SSD1306 OLED display communicates with the microcontroller through the $I^2C$ protocol using SDA and SCL lines.

The ESP32-S3 coordinates audio acquisition, voice command detection, and output control through the program's execution sequence. When a predefined command is detected, the controller updates the OLED display and activates the LED indicator. If no valid command is detected, the system continues capturing audio.

Proper synchronization between audio sampling, processing, and output updates helps reduce delays and improves system reliability. Local communication also minimizes dependence on external networks, supporting efficient and responsive voice activation.

# CHAPTER - 5

# Results and Discussion

## 5.1 Experimental Setup

The experimental setup of the Low Latency and Efficient Voice Activator for Edge Devices is designed to evaluate the performance and functionality of the proposed voice activation system. The setup consists of an ESP32-S3 development board, an INMP441 microphone, an SSD1306 OLED display, an LED indicator, connecting wires, and a USB power supply.

The ESP32-S3 is programmed using the Arduino IDE to capture audio data, process voice input, and detect predefined voice commands using a compatible voice detection algorithm. The microphone is connected through the $I^2S$ interface, while the OLED display communicates through the $I^2C$ interface. The LED provides a visual indication when a valid command is detected.

During the experiment, the system is powered on and initialized. Predefined voice commands are spoken near the microphone under different environmental conditions. The detected commands and system status are observed on the OLED display, and the LED response is monitored.

The system performance is evaluated using the following parameters:

1. Response Latency: Time taken to respond after a voice command is detected.
2. Detection Accuracy: Percentage of correctly detected commands.
3. Reliability: Consistency of operation across repeated trials.
4. Power Consumption: Electrical energy used during operation.
5. Resource Utilization: Memory and processing resources consumed by the system.

The experimental results help determine whether the system achieves its objectives of low-latency voice activation and efficient local processing. Actual measurements should be recorded during testing to validate the system's performance.

### 5.2 Output Results (Simulation Screenshots, Measured Readings, Waveforms)

The output results of the Low Latency and Efficient Voice Activator for Edge Devices are used to verify the functionality and performance of the proposed system. The ESP32-S3 processes audio signals captured by the INMP441 microphone and displays the system status on the SSD1306 OLED display. When a predefined voice command is detected, the LED indicator is activated.

1. Simulation and Hardware Output

The expected output messages displayed on the OLED include:

- System Initializing
- System Ready
- Listening...
- Command Detected
- Command Not Recognized

### 5.3 Performance Analysis

The performance analysis of the Low Latency and Efficient Voice Activator for Edge Devices evaluates the system in terms of response time, voice detection accuracy, power consumption, and resource utilization. The ESP32-S3 processes voice input locally, reducing dependence on cloud servers and network communication.

1. Response Latency: Response latency is the time between detecting a voice command and generating the corresponding output. Local processing can reduce communication delays and improve responsiveness.

2. Voice Detection Accuracy: Accuracy measures how correctly the system identifies predefined voice commands. Performance may vary depending on microphone quality, background noise, speaking volume, and the detection algorithm.

3. Power Consumption: The system uses an embedded microcontroller and peripheral devices. Power consumption depends on the processing workload, microphone, OLED display, LED, and operating mode.

4. Memory Utilization: The ESP32-S3 has limited memory compared with desktop computers. Lightweight voice detection algorithms help optimize RAM and flash memory usage.

5. Reliability: Reliability is evaluated by repeating voice command tests under different environmental conditions and observing whether the system responds consistently.

6. Overall Efficiency: Local voice processing reduces internet dependency and supports privacy-conscious operation. However, actual latency, accuracy, and energy efficiency must be established through experimental measurements.

In conclusion, the system is designed to provide fast and efficient voice activation on edge hardware. Its performance should be validated using measured response times, detection accuracy, power consumption, and memory utilization3 r5ather than assumed values.

# Chapter 6:

# Conclusion and Future Scope

## 6.1 Conclusion

TheLowLatency and Efficient Voice Activator for Edge Devices project focuses on developing a responsive, cost-effective, and energy-efficient voice activation system using edge computing technology. The proposed system uses the ESP32-S3 microcontroller, INMP441 microphone, SSD1306 OLED display, and LED indicator to capture voice input, detect predefined commands, and generate appropriate outputs.

By performing voice processing locally, the system reduces dependence on cloud servers and continuous internet connectivity. The use of lightweight voice detection techniques can help minimize response latency and optimize computational resources. The OLED display provides visual feedback, while the LED indicates successful command detection.

The project demonstrates the potential of integrating embedded systems, digital audio processing, and voice recognition techniques to enable convenient hands-free control. Potential applications include smart home automation, robotics, industrial control, and assistive technologies.

Overall, the proposed system provides a foundation for developing intelligent voice-controlled edge devices. Further testing is required to quantify its detection accuracy, response time, power consumption, and reliability under different environmental conditions. Future improvements can include advanced keyword spotting models, better noise suppression, and support for additional voice commands.

## 6.2 Future Scope

TheLow Latency and Efficient Voice Activator for Edge Devices project offers several opportunities for future development and improvement. By integrating advanced voice recognition techniques and optimized embedded hardware, the system can be enhanced to provide faster, more accurate, and more reliable voice-controlled operations.

1. Advanced Voice Recognition: Lightweight machine learning models can be integrated to improve keyword detection accuracy and support a wider range of voice commands.
2. Noise Reduction: Advanced digital signal processing techniques can be implemented to improve voice detection in noisy environments.
3. Multilingual Support: The system can be extended to recognize commands in multiple languages, making it accessible to a wider range of users.
4. Smart Home Automation: Voice commands can be used to control lights, fans, appliances, and other compatible devices through suitable control circuits.
5. Robotics Integration: The system can be connected to robotic platforms to support voice-controlled movement and task execution.
6. Power Optimization: Low-power operating modes and efficient algorithms can be used to reduce energy consumption in battery-powered applications.
7. Improved Security and Privacy: Local voice processing can be combined with secure command validation to reduce unauthorized device activation.
8. Performance Optimization: Future work can focus on reducing response latency, memory usage, and computational overhead while maintaining reliable detection accuracy.

In conclusion, these improvements can expand the practical applications of the project in smart homes, healthcare assistance, industrial automation, and intelligent embedded systems. The system can evolve into a more capable, energy-efficient, and reliable voice activation platform for next-generation edge devices.

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